Stability Analysis and Numerical Simulations

Rosenzweig–MacArthur model for lynx and hare

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1 Equilibria, Stability and Hopf Bifurcation

Throughout this section we analyse the dimensional Rosenzweig–MacArthur model

$$\frac{dH}{dt} = f(H,P) = rH\Big(1-\frac{H}{K}\Big) - \frac{aHP}{b+H}, \qquad \frac{dP}{dt} = g(H,P) = \frac{eaHP}{b+H} - mP, \tag{1}$$

with all six parameters r, K, a, b, e, m strictly positive, as defined in the model section. Every stability statement below is tied either to a derivation given here, to a cited theorem, or to a numerical check; Table 3 at the end of the section records which is which.

1.1 Well-posedness: invariance and boundedness

Lemma 1. *The closed quadrant $\{H \geq 0,\, P \geq 0\}$, both coordinate axes, and the open quadrant are invariant under* (1)*. Every solution starting in the quadrant is bounded, with*

$$\limsup_{t\to\infty} H(t) \leq K, \qquad \limsup_{t\to\infty} \big(eH(t) + P(t)\big) \leq \frac{eK(r+m)^2}{4rm}.$$

Proof. Both right-hand sides factor as $H\cdot(\cdot)$ and $P\cdot(\cdot)$, so $H=0$ and $P=0$ are invariant lines and, by uniqueness of solutions, no orbit can cross them. Since $\dot H \leq rH(1-H/K)$, comparison with the logistic equation gives $\limsup H \leq K$. For $W = eH + P$ the predation terms cancel: $\dot W = erH(1-H/K) - mP = eH\big(r(1-H/K)+m\big) - mW$. The quadratic $H\big(r(1-H/K)+m\big)$ is maximised at $H = K(r+m)/(2r)$ with value $K(r+m)^2/(4r)$, so $\dot W \leq eK(r+m)^2/(4r) - mW$ and the bound follows. □

Boundedness is what later allows the Poincaré–Bendixson theorem to be used.

1.2 Equilibria and biological feasibility

Setting $f = g = 0$:

- $P = 0$ in $g = 0$ leaves $rH(1-H/K) = 0$, giving the *origin* $E_0 = (0,0)$ and the *prey-only* equilibrium $E_1 = (K, 0)$.
- $P \neq 0$ requires $eaH/(b+H) = m$, i.e.

$$H^* = \frac{mb}{ea-m}, \tag{2}$$

and then $f = 0$ with $H = H^* \neq 0$ gives

$$P^* = \frac{r}{a}\Big(1-\frac{H^*}{K}\Big)(b+H^*) = \frac{ebr\big(K(ea-m)-bm\big)}{K(ea-m)^2}. \tag{3}$$

This is the *coexistence* equilibrium $E^* = (H^*, P^*)$.

Proposition 1 (Feasibility). *E_0 and E_1 always lie in the closed quadrant. E^* lies in the open quadrant if and only if*

$$ea > m \quad \textit{and} \quad H^* < K \qquad \Longleftrightarrow \qquad m < \frac{eaK}{b+K}.$$

Proof. $H^* > 0$ iff $ea > m$. Given this, $P^* > 0$ iff $1 - H^*/K > 0$. Rearranging $mb/(ea-m) < K$ gives $m(b+K) < eaK$. □

Biologically, ea is the predator's maximal per-capita birth rate (attained when prey is abundant), so $ea > m$ says lynx can outgrow their mortality at all; $H^* < K$ says the hare level lynx need to break even is actually reachable by the prey on its own. Note that H^* does not depend on r or K: the prey level at coexistence is set entirely by the predator.

1.3 Jacobian

Differentiating (1),

$$J(H,P) = \begin{pmatrix} r\Big(1 - \dfrac{2H}{K}\Big) - \dfrac{abP}{(b+H)^2} & -\dfrac{aH}{b+H} \\[2ex] \dfrac{eabP}{(b+H)^2} & \dfrac{eaH}{b+H} - m \end{pmatrix}. \tag{4}$$

For a planar system, with $\tau = \operatorname{tr} J$ and $\Delta = \det J$, an equilibrium is a saddle if $\Delta < 0$; is asymptotically stable if $\Delta > 0$, $\tau < 0$; and is unstable if $\Delta > 0$, $\tau > 0$. It is a focus when $\tau^2 < 4\Delta$ and a node when $\tau^2 > 4\Delta$.

1.4 Local stability of each equilibrium

Origin. $J(E_0) = \operatorname{diag}(r, -m)$, with eigenvalues $r > 0$ and $-m < 0$. E_0 is *always a saddle*: the P-axis is its stable manifold (lynx alone die out) and the H-axis its unstable manifold (hares alone grow). Joint extinction is never an attractor.

Prey-only equilibrium.

$$J(E_1) = \begin{pmatrix} -r & -\dfrac{aK}{b+K} \\[2ex] 0 & \dfrac{eaK}{b+K} - m \end{pmatrix}, \qquad \lambda_1 = -r, \quad \lambda_2 = \frac{eaK}{b+K} - m.$$

E_1 is a stable node when $\lambda_2 < 0$, i.e. exactly when E^* is *not* feasible, and a saddle when $\lambda_2 > 0$, i.e. exactly when E^* is feasible. At $K = H^*$ the two equilibria coincide and exchange stability (a transcritical bifurcation). λ_2 is the invasion rate of a rare lynx population into a hare population at carrying capacity.

Coexistence equilibrium. At E^* the predator equation gives $eaH^*/(b+H^*) = m$, so $J_{22} = 0$. Using the prey nullcline, $aP^*/(b+H^*) = r(1 - H^*/K)$, to eliminate P^* from J_{11}:

$$J_{11} = r\Big(1 - \frac{2H^*}{K}\Big) - \frac{b}{b+H^*}\, r\Big(1 - \frac{H^*}{K}\Big) = \frac{rH^*\,(K - b - 2H^*)}{K\,(b+H^*)}.$$

Hence

$$\tau(E^*) = \frac{rH^*}{K(b+H^*)}\,\big(K - K_H\big), \qquad K_H := b + 2H^* = \frac{b\,(ea+m)}{ea-m}, \tag{5}$$

$$\Delta(E^*) = -J_{12}J_{21} = \frac{aH^*}{b+H^*} \cdot \frac{eabP^*}{(b+H^*)^2} = \frac{rmb\,(K - H^*)}{K(b+H^*)}. \tag{6}$$

Theorem 1 (Local stability of E^*)**.** *Suppose E^* is feasible ($H^* < K$). Then $\Delta(E^*) > 0$, so E^* is never a saddle, and*

$$E^* \text{ is locally asymptotically stable if } K < K_H, \qquad E^* \text{ is unstable (a repeller) if } K > K_H.$$

Proof. $\Delta > 0$ follows from (6) since $K > H^*$. The sign of τ is the sign of $K - K_H$ by (5). Apply the trace–determinant classification. □

Remark 1. The intrinsic hare growth rate r appears in τ and Δ only as a positive prefactor. It therefore sets how fast trajectories spiral, but has *no effect* on whether E^* is stable. Near feasibility (K just above H^*), $\Delta \to 0$ while $\tau < 0$, so E^* is a stable node; as K increases it becomes a stable focus once $\tau^2 < 4\Delta$ (at $K \approx 42.5$ for the illustrative parameters of §1.8), so convergence is oscillatory well before the Hopf point.

Table 1: Equilibria of (1) and their local stability.

Equilibrium	Feasible when	Eigenvalues / invariants	Type
$E_0 = (0, 0)$	always	$r,\ -m$	saddle
$E_1 = (K, 0)$	always	$-r,\ \frac{eaK}{b+K} - m$	stable node if $K < H^*$ or $ea \le m$; saddle if $K > H^*$
$E^* = (H^*, P^*)$	$ea > m,\ H^* < K$	$\tau \propto K - K_H,\ \Delta > 0$	stable if $K < K_H$; unstable if $K > K_H$

1.5 Nullclines and the phase plane

Prey nullclines ($\dot{H} = 0$): the P-axis $H = 0$, and the parabola

$$P = \phi(H) := \frac{r}{a}\Big(1 - \frac{H}{K}\Big)(b + H), \tag{7}$$

which is concave, vanishes at $H = -b$ and $H = K$, cuts the P-axis at rb/a, and peaks at

$$H_{\text{peak}} = \frac{K - b}{2}, \qquad \phi(H_{\text{peak}}) = \frac{r(K + b)^2}{4aK}.$$

$\dot{H} > 0$ below the parabola and $\dot{H} < 0$ above it (for $H > 0$).

Predator nullclines ($\dot{P} = 0$): the H-axis $P = 0$, and the vertical line $H = H^*$. $\dot{P} > 0$ to the right of $H = H^*$ (enough prey) and $\dot{P} < 0$ to the left.

Flow. The two interior nullclines divide the open quadrant into four regions. Starting below the parabola to the right of H^*, both species increase; crossing the parabola hares start to fall while lynx still rise; crossing $H = H^*$ both fall; crossing back below the parabola hares recover while lynx are still falling. The flow therefore circulates *counter-clockwise* around E^* in the (H, P) plane, and lynx peaks lag hare peaks. Near the Hopf point the lag is a quarter-period; it grows with K (numerically, 2.5 to 4.5 years for regimes D–F below).

Graphical stability criterion. Since $K_H = b + 2H^*$ is equivalent to $H^* = (K - b)/2 = H_{\text{peak}}$, Theorem 1 says:

E^* is stable if the predator nullcline cuts the prey parabola to the right of its hump, and unstable if it cuts to the left.

This is the original Rosenzweig–MacArthur graphical criterion [6]. It also gives the sign of J_{11} directly: $J_{11} = a\,H^*\phi'(H^*)/(b + H^*)$, so a rising nullcline (left of the hump) means local prey self-amplification ($J_{11} > 0$). Figure 1 shows both cases.

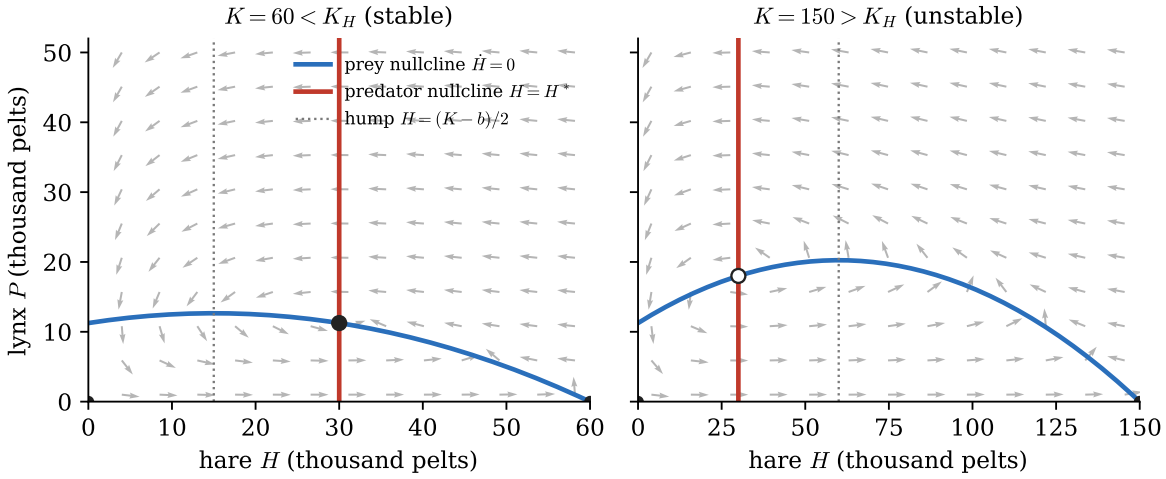


Figure 1: Nullclines and direction field for the illustrative parameters ($r = 1.5$, $a = 4$, $b = 30$, $e = 0.4$, $m = 0.8$; $H^* = 30$, $K_H = 90$). Left: $K = 60$; the predator nullcline meets the parabola right of its hump and E^* (filled) is a stable focus. Right: $K = 150$; the intersection is left of the hump and E^* (open) is unstable. Small dots mark E_0 and E_1.

1.6 Hopf bifurcation with K as the parameter

We fix r, a, b, e, m with $ea > m$ and vary K over (H^*, ∞). Note H^* is independent of K, while P^* increases with K.

Theorem 2 (Hopf bifurcation). *At $K = K_H = b + 2H^*$ the coexistence equilibrium undergoes a Hopf bifurcation:*

(i) $\tau(E^*) = 0$ *and* $\Delta(E^*) = \dfrac{rmb\,(K_H - H^*)}{K_H(b + H^*)} = \dfrac{rmb}{K_H} > 0$, *so $J(E^*)$ has eigenvalues $\pm i\omega_0$ with*

$$\omega_0 = \sqrt{\frac{rmb}{b + 2H^*}}, \qquad T_0 = \frac{2\pi}{\omega_0};$$

(ii) (transversality) the real part of the eigenvalues, $\tau/2$, crosses zero with nonzero speed:

$$\frac{d}{dK}\left(\frac{\tau}{2}\right)\Big|_{K=K_H} = \frac{rH^*}{2(b + H^*)(b + 2H^*)} > 0.$$

Proof. (i) Substitute $K = K_H$ into (5) and (6); $K_H - H^* = b + H^*$ simplifies Δ. (ii) Differentiate (5) in K: $\frac{d}{dK}\frac{K - K_H}{K} = \frac{K_H}{K^2}$, which equals $1/K_H$ at $K = K_H$. All symbolic steps were checked with SymPy (`scripts/joel_symbolic_derivation.py`) and the closed forms are tested against the shared model code (`tests/test_stability.py`). □

Conditions (i)–(ii) establish that a pair of complex eigenvalues crosses the imaginary axis. To conclude that a small periodic orbit is born, and on which side, the Hopf theorem also needs the first Lyapunov coefficient ℓ_1 to be nonzero [4]: $\ell_1 < 0$ gives a *supercritical* bifurcation (a stable cycle for $K > K_H$) and $\ell_1 > 0$ a subcritical one.

What is analytic and what is numerical. We separate four statements.

1. **Analytic (derived here):** the threshold K_H, conditions (i)–(ii), and the exchange of stability at E^*.

2. **Analytic (derived here): existence of a periodic orbit for $K > K_H$.** Take any initial condition in the open quadrant. By Lemma 1 its ω-limit set Ω is nonempty and compact. Ω cannot contain E^*, which is a repeller. Ω cannot be $\{E_1\}$, because the stable manifold of the saddle E_1 is the H-axis. If Ω contained E_1 together with other points, it would contain a point of E_1's stable manifold on the H-axis and hence the whole axis orbit through it, which tends backward to E_0; repeating the argument at E_0 would put a point of the P-axis in Ω, whose backward orbit is unbounded, contradicting compactness. So Ω contains no equilibria, and the Poincaré–Bendixson theorem [8] makes Ω a periodic orbit.

3. **Cited theorems:** Cheng [1] proved the limit cycle is *unique* for $K > K_H$; since every interior orbit other than E^* tends to a periodic orbit, that unique cycle attracts them all and is orbitally stable. Hsu [3] proved that a locally stable E^* is *globally* stable in the open quadrant whenever the prey nullcline is concave, which our parabola (7) is, so E^* attracts every interior orbit for $H^* < K < K_H$. We use these results without reproducing their proofs.

4. **Numerical (for our parameter set only):** evaluating Kuznetsov's formula for ℓ_1 at $K = K_H$ gives $\ell_1 \approx -4.0 \times 10^{-4} < 0$ (and -1.3×10^{-3} for a second set $r = 0.7$, $m = 0.5$, $b = 20$), so the bifurcation is supercritical, consistent with the cited uniqueness result. Simulations confirm the supercritical signature: the post-transient cycle half-amplitude in H scales as $(K - K_H)^{1/2}$ (fitted log–log slope 0.507 for $K - K_H \in [0.5, 8]$), and the period tends to T_0.

None of this says that the *real* hare–lynx data sit on a limit cycle. That depends on whether fitted parameters satisfy $K > K_H$, and with about two oscillations in 1900–1920 a slowly damped oscillation (K slightly below K_H) can fit the same data. That question belongs to the fitting sections and must be answered there with uncertainty.

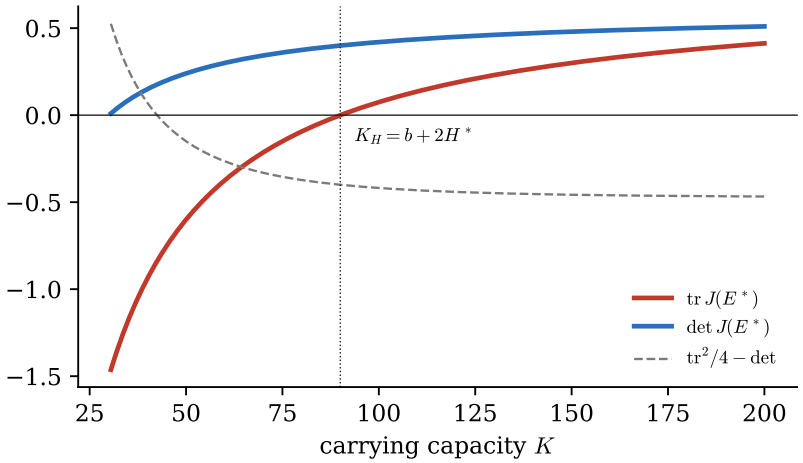


Figure 2: Trace and determinant of $J(E^*)$ as functions of K (illustrative parameters). The determinant stays positive for all feasible $K > H^* = 30$, while the trace changes sign at $K_H = 90$. The dashed curve is negative (focus) for $K > 42.5$.

1.7 The paradox of enrichment

Increasing K (enriching the hares' environment) leaves H^* unchanged and only raises P^*: every extra unit of carrying capacity is passed to the predator. Graphically, the prey parabola's hump

$(K-b)/2$ moves right while the predator nullcline stays at H^*, so for large enough K the intersection falls left of the hump and the equilibrium destabilises. Past K_H the system settles on a limit cycle whose amplitude grows with K. For our illustrative parameters, at $K = 150$ the hare minimum on the cycle is about 1.1 thousand, compared with $H^* = 30$. In a finite, stochastic population such deep troughs bring a real risk of extinction. Enrichment thus *destabilises* the system, the paradox named by Rosenzweig [5].

The mechanism is the saturating Holling type II response. At low H each lynx's consumption rises almost linearly with prey, which damps fluctuations; near and above b it saturates, so per-capita predation risk $aP/(b+H)$ *falls* as hares increase. That positive feedback on prey growth is exactly the term making $J_{11} > 0$ left of the hump.

Link to Lotka–Volterra. In the classical model ($K = \infty$, linear response) the analogous Jacobian at coexistence has $\tau \equiv 0$ for every parameter value, so it sits permanently on the Hopf boundary. That is why its neutral cycles are structurally unstable: any small change in the model's form moves it to one side or the other. Rosenzweig–MacArthur makes the side explicit through the sign of $K - K_H$.

1.8 Illustrative regimes and numerical validation targets

These parameters are *illustrative, not fitted*: H and P in thousands of pelts, t in years, with $r = 1.5,\ a = 4,\ b = 30,\ e = 0.4,\ m = 0.8$. Then $ea = 1.6 = 2m$, so $H^* = b = 30$, $K_H = 90$, $\omega_0 = \sqrt{0.4} \approx 0.632$ and $T_0 \approx 9.94$ years, close to the observed decadal cycle. Table 2 gives the expected behaviour on each side of the threshold; the simulation section reproduces these with RK45 (`rtol`$= 10^{-10}$, `atol`$= 10^{-12}$, 3000-year burn-in, start $(20, 5)$).

Table 2: Regimes in K (other parameters fixed). Eigenvalues are of $J(E^*)$, except in A where they are of $J(E_1)$.

	K	Linear prediction	Eigenvalues	Long-run behaviour (simulated)
A	25	E^* infeasible ($K < H^*$)	$-1.5,\ -0.073$	lynx die out; $\to (25, 0)$
B	60	E^* stable focus	$-0.188 \pm 0.515i$	$\to$ (30, 11.25), e-fold ≈ 5 yr
C	85	just below K_H	$-0.022 \pm 0.623i$	$\to$ (30, 14.56), e-fold ≈ 45 yr
D	90	Hopf point	$\pm 0.632i$	very slow (algebraic) decay
E	95	just above K_H	$+0.020 \pm 0.640i$	cycle: $H \in [15.8, 50.9]$, $P \in [9.8, 20.8]$, $T = 10.25$
F	150	well above K_H	$+0.150 \pm 0.676i$	cycle: $H \in [1.1, 134.3]$, $P \in [1.7, 41.2]$, $T = 13.91$

Two cautions for the numerics. At D the equilibrium is a weak focus ($\ell_1 < 0$), so oscillations decay only like $t^{-1/2}$; after 3000 years the residual amplitude in H is still about 1.5, which is expected and is not a sign of a cycle. The cycle at E is already large because the amplitude coefficient is large ($A_H \approx 7.8\sqrt{K - K_H}$), so "small amplitude" only holds within about one unit of K_H.

1.9 Sensitivity of the threshold

Because $K_H = b(ea+m)/(ea-m)$ is explicit, sensitivity can be read off exactly. The elasticities (percentage change in K_H per percentage change in a parameter) are

$$\frac{\partial \ln K_H}{\partial \ln b} = 1, \qquad \frac{\partial \ln K_H}{\partial \ln m} = -\frac{\partial \ln K_H}{\partial \ln a} = -\frac{\partial \ln K_H}{\partial \ln e} = \frac{2\,ea\,m}{(ea)^2 - m^2}, \qquad \frac{\partial \ln K_H}{\partial \ln r} = 0.$$

At the illustrative values ($ea = 2m$) the middle elasticity is $4/3$: a 10% rise in lynx mortality raises the threshold by about 13%, and so stabilises the system. The elasticity blows up as $m \to ea$, where the predator is barely viable and $K_H \to \infty$. Weak predators stabilise.

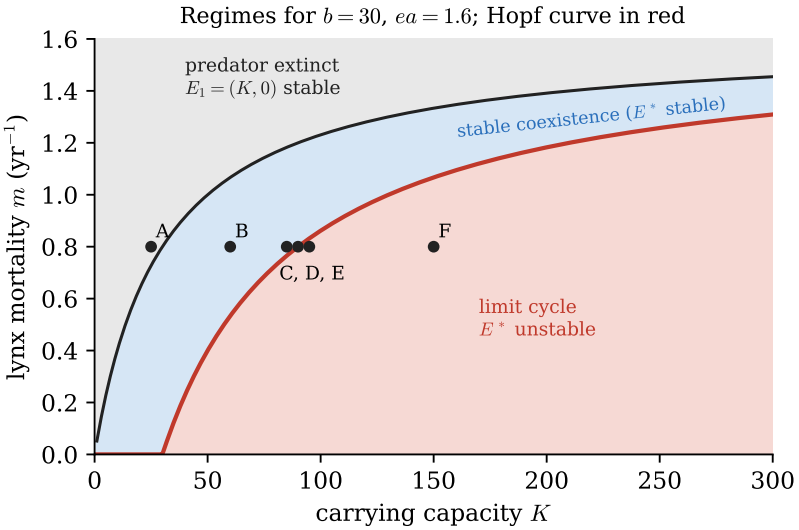


Figure 3: Analytic regimes in the (K, m) plane for $b = 30$, $ea = 1.6$. Black curve: $m = eaK/(b+K)$ (predator persistence boundary). Red curve: $m = ea(K-b)/(K+b)$ (Hopf curve). Dots mark regimes A–F of Table 2.

Equivalently, for fixed K the regimes are bands in m (Figure 3):

$$\underbrace{m < \frac{ea(K-b)}{K+b}}_{\text{limit cycle}} \qquad \underbrace{\frac{ea(K-b)}{K+b} < m < \frac{eaK}{K+b}}_{\text{stable coexistence}} \qquad \underbrace{m > \frac{eaK}{K+b}}_{\text{lynx extinct}}$$

A useful single number for any fitted parameter set is the *stability margin* $\sigma = K/K_H$: $\sigma < 1$ means damped oscillations, $\sigma > 1$ a limit cycle. Reporting σ (with its spread over fits) is the cleanest way to say whether the data support sustained cycling.

Two structural facts matter for inference. First, a and e enter H^*, K_H and the stability conditions only through the product ea. Second, if lynx are measured only up to an unknown multiplicative factor c (as pelt counts are), the map $(P, a, e) \mapsto (cP,\ a/c,\ ec)$ leaves (1) unchanged (checked symbolically), so a and e are *not separately identifiable* from index data. Fitting should fix one of them, or fit ea and a scale, rather than report both as biological rates.

1.10 Dimensionless form of the results

With the scaling $x = H/K$, $y = aP/(rK)$, $s = rt$ (to be reconciled with the frozen scaling in the model section), (1) becomes

$$x' = x(1-x) - \frac{xy}{\beta + x}, \qquad y' = y\Big(\frac{\alpha x}{\beta + x} - \delta\Big), \qquad \beta = \frac{b}{K},\ \alpha = \frac{ea}{r},\ \delta = \frac{m}{r}.$$

Then $x^* = \beta\delta/(\alpha - \delta)$, coexistence requires $\alpha > \delta$ and $x^* < 1$, and the Hopf condition $K = K_H$ reads $\beta(\alpha + \delta)/(\alpha - \delta) = 1$, i.e. $\beta + 2x^* = 1$. Only three combinations of the six parameters control the qualitative dynamics.

Table 3: Status of each stability claim in this section.

Claim	Evidence
Equilibria, feasibility, local stability of E_0, E_1, E^*	Derived here; SymPy-checked
Hopf threshold $K_H = b + 2H^*$, $\Delta > 0$, transversality	Derived here; SymPy-checked
A periodic orbit exists for every $K > K_H$	Derived here (Poincaré–Bendixson)
Cycle is unique and attracting for $K > K_H$	Cited theorem [1]
E^* globally stable for $H^* < K < K_H$	Cited theorem [3]
Supercritical Hopf ($\ell_1 < 0$)	Numerical evaluation of analytic formula, two parameter sets
$\sqrt{K - K_H}$ amplitude law; periods, amplitudes, lags	Numerical (RK45), illustrative parameters
Real hare–lynx data lie on a limit cycle	*Not established here*; requires $\sigma > 1$ at fitted values with uncertainty

2 Numerical Simulations

Section 1 locates the Hopf threshold $K_H = b + 2H^*$ analytically. Linearisation is, however, a local statement about the equilibrium at a single parameter value. This section tests those predictions numerically and measures what linear analysis cannot give: the long-run behaviour away from E^*, the size and period of the cycle, and the shape of the bifurcation. Every trajectory is computed with the group's shared right-hand side of (1).

2.1 Method

Parameters. We use the illustrative, unfitted values of Section 1: $r = 1.5$, $a = 4$, $b = 30$, $e = 0.4$, $m = 0.8$ (H, P in thousands of pelts, t in years), for which $H^* = 30$, $K_H = 90$ and $T_0 = 2\pi/\omega_0 \approx 9.94$ years. The carrying capacity K is varied.

Solver. Equations (1) are integrated with the adaptive Dormand–Prince Runge–Kutta pair (RK45) in `scipy.integrate.solve_ivp` [2, 7]. At each step a fourth- and a fifth-order estimate are formed from the same seven slope evaluations; their difference estimates the local error, and the step is rejected and reduced whenever this exceeds `rtol` $|y|$ + `atol`. The project uses two settings of this solver for two different jobs. Fitting and the 1900–1920 forecasts of the historical sections use `model.solve` with `rtol` $= 10^{-8}$, `atol` $= 10^{-10}$ and a maximum step of 0.1 years, identical for both fitting methods. The long-run simulations in this section run for thousands of years, so they use tighter tolerances, `rtol` $= 10^{-10}$ and `atol` $= 10^{-12}$, without a step cap. Both call the same shared right-hand side. The illustrative runs start from $(H_0, P_0) = (20, 5)$; all values are in `configs/simulations.json`.

Long-run behaviour. To describe the attractor we integrate for a burn-in period T_{burn}, discard it, and record the next 300–400 years. On the recorded window we report the range of H and P; if H still oscillates, the period is the mean interval between successive hare maxima and the lag is the mean delay from a hare maximum to the next lynx maximum. We take $T_{\text{burn}} = 3000$ years, increased to 20,000 years when $|K - K_H| < 1$, where the equilibrium is a weak focus and transients decay very slowly.

2.2 Solver accuracy

The classical Lotka–Volterra system $\dot{H} = \alpha H - \beta HP$, $\dot{P} = \delta HP - \gamma P$ conserves $V = \delta H - \gamma \ln H + \beta P - \alpha \ln P$, so any drift in V along a computed orbit is numerical error. Over 100 time units RK45 keeps $|V(t) - V(0)|$ below 4×10^{-10}, whereas explicit Euler with step $h = 0.01$ drifts by 0.15 and its orbit spirals outward (Figure 4), creating oscillatory growth the model does not have. For Rosenzweig–MacArthur at $K = 150$, tightening both tolerances a hundredfold changes the solution by a relative 8×10^{-10}, and all states remain positive, as Lemma 1 requires.

2.3 Reproduction of the analytic regimes

Table 4 reproduces each validation target of Section 1 (3000-year burn-in throughout). Regime D lies exactly on K_H: the residual oscillation there is a slowly decaying transient (decay is only algebraic at a weak focus), not a cycle.

Figure 5 shows four of these regimes. Below the threshold (B, C) the predator nullcline meets the prey hump to the right of its peak and trajectories spiral into E^*; at $K = 85$ the trace is only -0.044, so the decay takes centuries. Above the threshold (E, F) the intersection lies left of the peak, E^* repels, and the orbit settles onto a sustained cycle that circulates counter-clockwise, with lynx peaks lagging hare peaks.

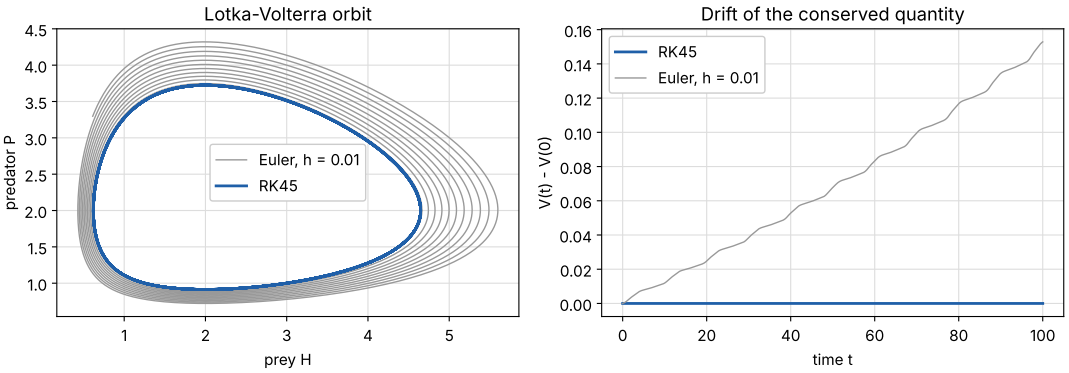


Figure 4: Solver check on Lotka–Volterra. RK45 preserves the conserved quantity V and closes the orbit; Euler ($h = 0.01$) drifts and spirals outward. Only an accurate solver can distinguish a genuine limit cycle from a numerical artefact.

Table 4: Simulated long-run behaviour for the regimes of Section 1. Ranges are over 400 years after the burn-in; period and lynx lag are in years.

	K	Eigenvalues	H range	P range	Period	Lag
A	25	$-1.5,\ -0.073$ (E_1)	25.0	0	–	–
B	60	$-0.188 \pm 0.515i$	30.0	11.25	–	–
C	85	$-0.022 \pm 0.623i$	30.0	14.56	–	–
D	90	$\pm 0.632i$	$[28.5, 31.6]$	$[14.51, 15.49]$	9.94	2.50
E	95	$+0.020 \pm 0.640i$	$[15.8, 50.9]$	$[9.81, 20.75]$	10.25	2.81
F	150	$+0.150 \pm 0.676i$	$[1.1, 134.3]$	$[1.69, 41.16]$	13.91	4.52

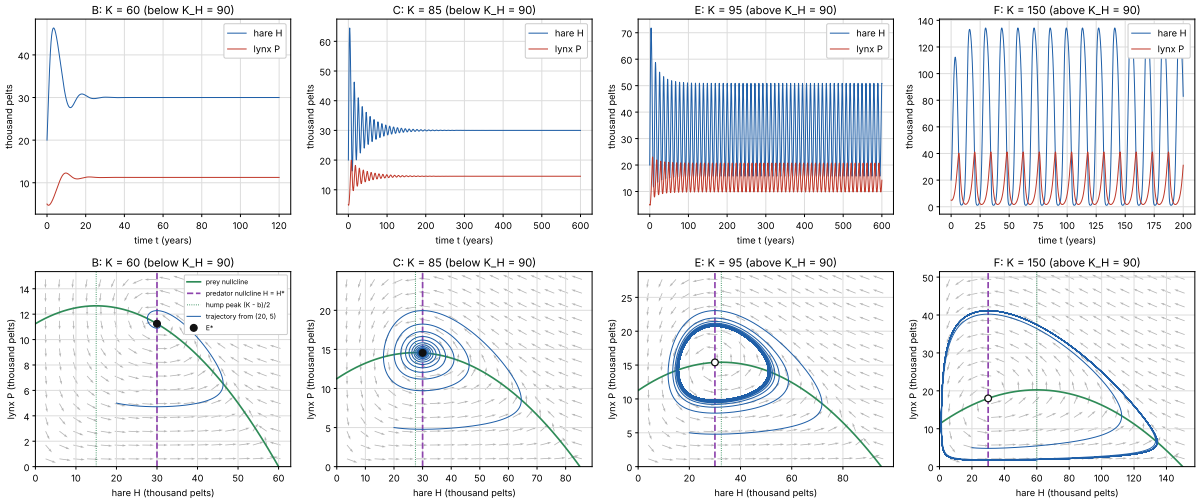


Figure 5: Time series (top) and phase portraits (bottom) from $(20, 5)$ for $K = 60, 85$ (below $K_H = 90$) and $K = 95, 150$ (above). Green: prey nullcline; dashed purple: predator nullcline $H = H^*$; dotted: hump peak $(K - b)/2$. Filled/open circle: stable/unstable E^*. Oscillations decay below K_H and persist above it.

2.4 The limit cycle attracts

For Lotka–Volterra every initial condition lies on its own closed orbit. At $K = 150$ we start one trajectory beside the unstable equilibrium and one far outside the cycle (Figure 6). The first spirals outward, the second inward, and their successive hare maxima converge to the same value, 134.277. The cycle is therefore orbitally stable, in agreement with the uniqueness result cited in Section 1 [1].

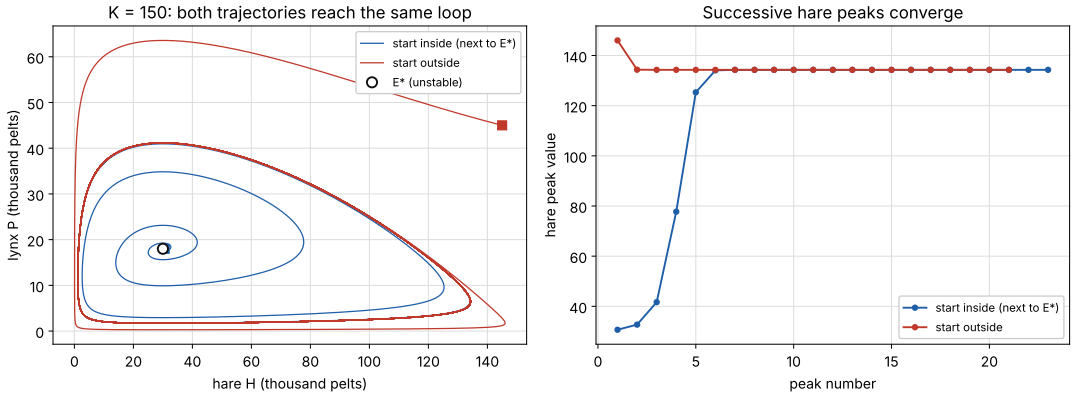


Figure 6: At $K = 150$, trajectories started inside and outside the cycle converge onto the same closed orbit (left); their successive hare maxima approach a common value (right).

2.5 Bifurcation in K

Root-finding on the numerically evaluated trace of $J(E^*)$ places the sign change at $K = 90.0000000000$, in agreement with $K_H = b + 2H^*$ and Figure 2; there the eigenvalues are $\pm 0.632i = \pm i\omega_0$, and they become complex (node to focus) at $K \approx 42.6$.

The numerical bifurcation diagram (Figure 7) records the post-transient maximum and minimum for 113 values of K between 32 and 200. For $K < 89$ the two coincide (largest spread 5×10^{-5}): the solution settles on E^*. At K_H the branches separate continuously from the equilibrium and the cycle grows with K. Just above the threshold its half-amplitude follows $A \approx 7.8\sqrt{K - K_H}$, with fitted log–log slope 0.507 for $0.5 \leq K - K_H \leq 8$ (Figure 8), and the period approaches $T_0 = 9.93$ years (9.97 years at $K - K_H = 0.5$). This square-root growth from zero amplitude is the signature of a supercritical Hopf bifurcation, consistent with the negative first Lyapunov coefficient of Section 1. As K increases further the cycle minima approach zero: at $K = 200$ the hare minimum is 0.12 thousand pelts, the paradox of enrichment [5].

2.6 Where the historical fits lie

We now evaluate the same quantities at every parameter set fitted to the 1900–1915 pelt series in the historical sections: 13 PINN runs (three seeds for each error model, the wider-bounds runs and the physics-weight run) and all 18 matched least-squares starts. Only saved parameters and the training file are read; the held-out years are not used. As a check that the parameters are interpreted exactly as in fitting, re-integrating the selected PINN fit with `model.solve` and the historical solver settings reproduces the saved 1900–1920 trajectory to within 5×10^{-12}. For each set we compute $\sigma = K/K_H$, the eigenvalues $\lambda = \mu \pm i\omega$ of $J(E^*)$, the e-folding time $1/|\mu|$ and the linear period $2\pi/\omega$. We also record the period and lynx lag of the first century simulated from the fitted 1900 state, and flag coefficients within 0.5% of a box limit. Five least-squares starts end in local minima with a training error at least 1.5 times the best start of their variant; they are shown but not interpreted. The complete table is `results/simulations/historical/fits.csv`.

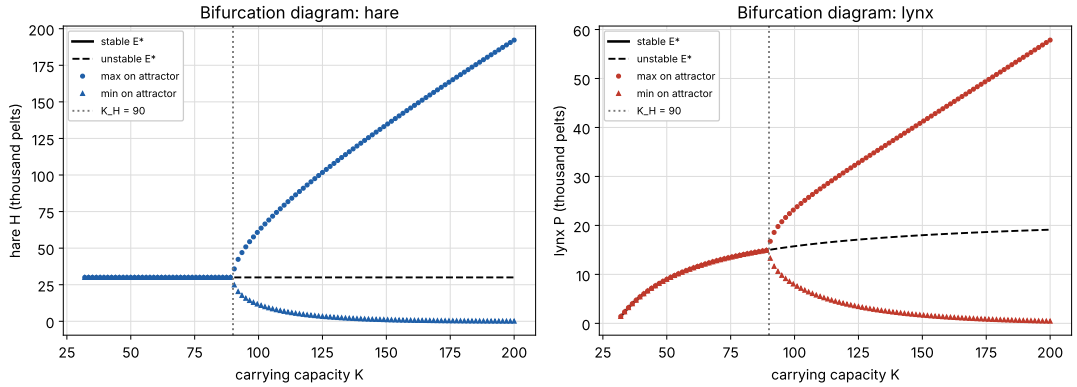


Figure 7: Numerical bifurcation diagram: maximum and minimum of H (left) and P (right) after the burn-in, against K. Solid/dashed black: stable/unstable E^*. A cycle is born at $K_H = 90$ with zero amplitude and its minima approach zero as K grows.

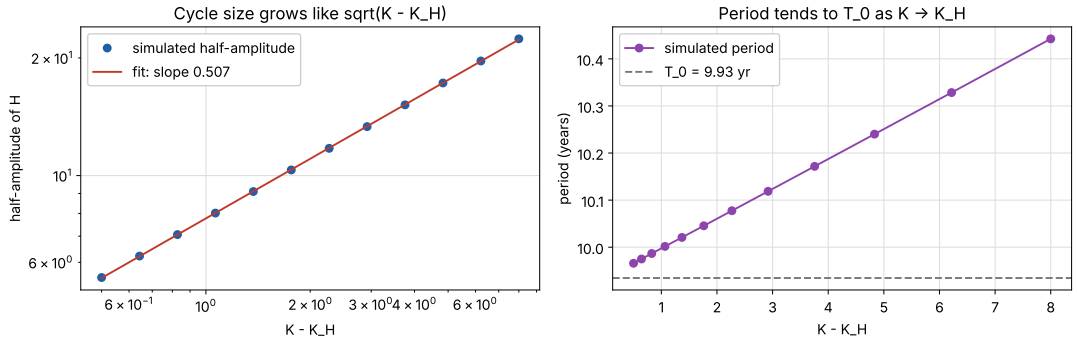


Figure 8: Left: half-amplitude of H against $K - K_H$ on log–log axes, with slope 0.507 (supercritical prediction 1/2). Right: cycle period, tending to $T_0 = 2\pi/\omega_0$.

Table 5: Selected historical fits (lowest training objective within each variant). $\mu \pm i\omega$: eigenvalues at E^*; periods and lag in years; "bounds": coefficients within 0.5% of their box limit.

Method	Variant	σ	$\mu \pm i\omega$	$1/\lvert\mu\rvert$	$2\pi/\omega$	Period, lag	Bounds
PINN	main	1.363	$+0.0079 \pm 0.712i$	127	8.8	10.7, 1.4	K, b upper
PINN	log errors	1.343	$+0.0079 \pm 0.767i$	127	8.2	9.9, 1.3	–
PINN	fitted H_0, P_0	1.347	$+0.0070 \pm 0.699i$	143	9.0	10.7, 1.4	b upper
PINN	wider bounds	1.480	$+0.0049 \pm 0.709i$	204	8.9	10.5, 1.3	b upper
PINN	physics $\times 10$	1.364	$+0.0091 \pm 0.687i$	111	9.1	11.3, 1.4	–
Least squares	main	1.067	$+0.0018 \pm 0.683i$	554	9.2	10.4, 1.6	a, b upper
Least squares	log errors	0.550	$-0.0190 \pm 0.680i$	53	9.2	9.4, 2.0	b upper
Least squares	fitted H_0, P_0	0.993	$-0.0002 \pm 0.669i$	5600	9.4	10.3, 1.7	b upper

Table 5 and Figure 9 show three findings. First, every well-fitting set reproduces the qualitative features of the data: an oscillation with a period of about 10 years and a lynx lag of 1.3–2 years. Second, these sets disagree about the side of the threshold. All PINN fits give σ between 1.32 and 1.48, whereas the least-squares fits give 1.07, 0.99 and 0.55 for the main, fitted-initial-state and log-error variants. Third, the half-saturation constant b lies on its upper box limit in nearly every fit, and widening the box moves the PINN estimate from $\sigma \approx 1.36$ to 1.48, so the reported σ is conditional on the bounds.

The reason the data cannot settle the question is the size of μ. In every well-fitting set $|\mu|$ is between 0.0002 and 0.019 per year, so oscillations grow or decay by a factor e only over 53 to 5600 years. Over the 16 training years the factor $e^{16\mu}$ lies between about 0.74 and 1.16, and a slowly damped focus and a slowly growing spiral produce almost the same two cycles. Figure 10 shows the bifurcation diagram in K for the two selected main fits, with all other coefficients held at their fitted values. For the least-squares fit the long-run cycle (H between 5.6 and 85 thousand pelts, period 10.8 years) resembles the observed range. For the PINN fit the long-run cycle is much larger (H between 0.1 and 247, period 18.5 years); the fitted trajectory over 1900–1920 is still an early, slowly growing part of that spiral.

We therefore do not conclude that the historical system lies on a stable limit cycle. The fitted model is close to the Hopf threshold, oscillates with the observed period and lag, and which side of the threshold it falls on depends on the fitting method, the error model and the bounds.

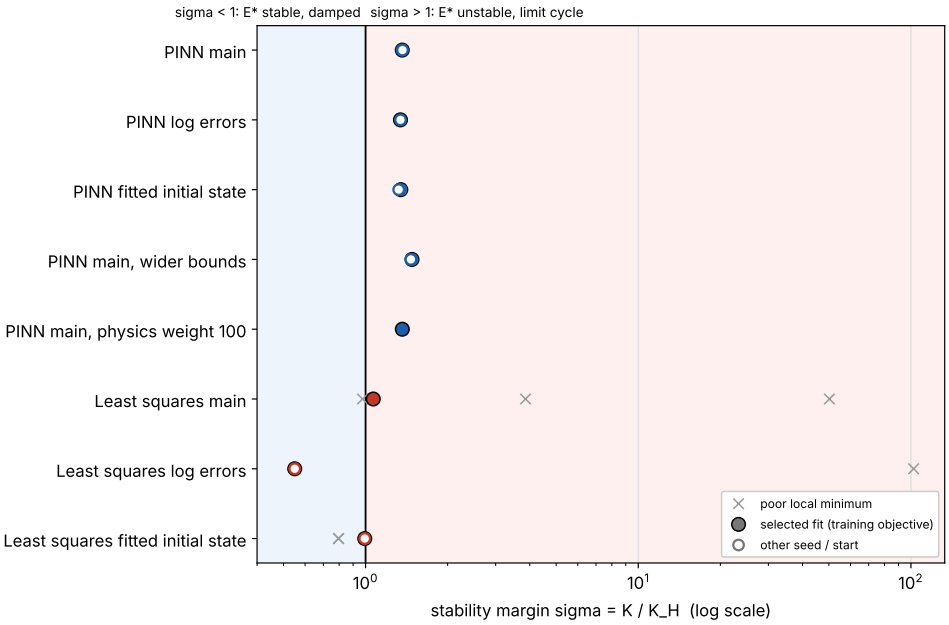


Figure 9: Stability margin $\sigma = K/K_H$ of every fitted parameter set (log scale). Filled: selected fit of each variant; open: other seeds or starts; crosses: least-squares local minima with at least 1.5 times the best training error. PINN fits lie at $\sigma \approx 1.3$–1.5; least-squares fits straddle $\sigma = 1$.

2.7 Scope

The illustrative simulations provide numerical evidence for one parameter set, not proofs. Near K_H the results depend on the burn-in length, which is why it is lengthened there. The historical placement uses the saved fits as given; their σ values inherit the bounds, the error model and the observation convention of the fitting sections, and the pelt series are abundance indices rather than population counts.

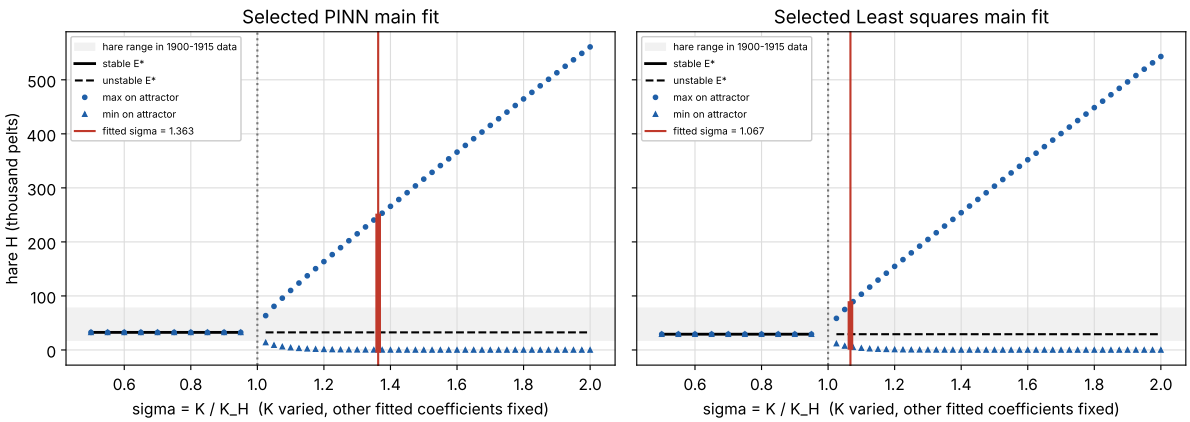


Figure 10: Bifurcation diagram in K for the selected PINN (left) and least-squares (right) main fits, plotted against $\sigma = K/K_H$ with the other fitted coefficients fixed. Red line: fitted σ; thick red bar: long-run hare range at the fitted parameters; grey band: hare range in the 1900–1915 training data. Near $\sigma = 1$ transients decay very slowly, so points there still carry residual oscillation after the burn-in (at most 60,000 years).

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